

Marmara University

Faculty of Engineering

PHYS 1103.4

PHYSICS LABORATORY-2024 FALL

EXPERIMENT 2

CONSTANT VELOCİTY and ACCELERATİON

Section Number: 17

Group Number: 5

Submitted to: Arş. Gör. Hüseyin YALTIRIK

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1. Purpose

In this experiment, the aim is to compare theoretical predictions and results of the experiment of objects which have constant velocity or acceleration. The experiment is repeated by changing the slope of the ground to see how it affects the acceleration.

2. Theory

A List of Materials

1) **Vernier Labquest Mini**
2) **Motion Detector**
3) **Moving Object**
4) **Ruler**

Two separate experiments will be conducted. First one will be constant velocity experiment and second one will be constant acceleration.

The constant velocity experiment is conducted to demonstrate that an object moves equal distances in equal amounts of time. To satisfy the conditions of constant velocity experiment, the ground must be a straight line. That's why, the ground's angle must be 0. The ground's angle should be measured by using protractor.

$$v = \frac{\Delta x}{\Delta t}$$

(v): (velocity)
(Delta x): Change in position (final position - initial position)
(Delta t): Time interval (final time - initial time)



1. Figure 1: Setup for experiment of constant velocity

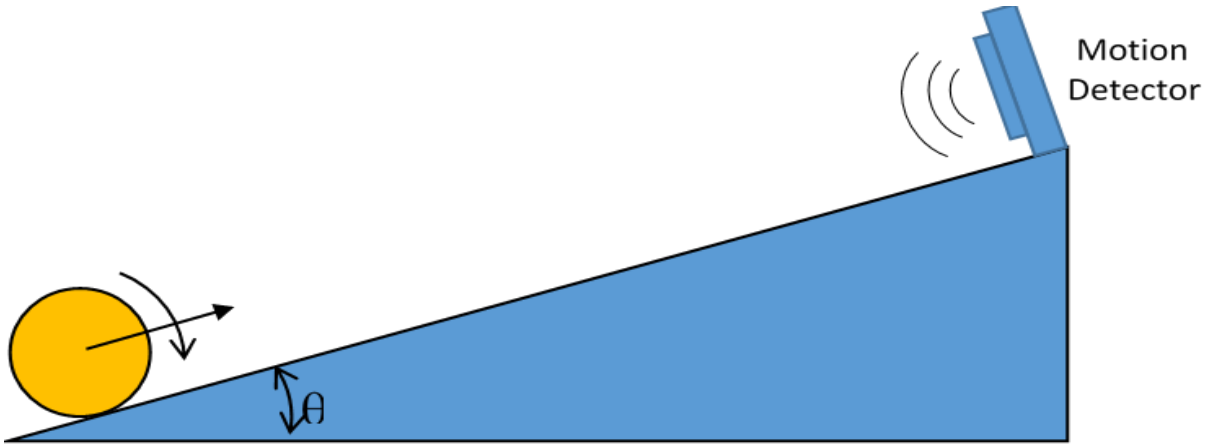
Acceleration is defined as the change of the velocity at a particular time. Constant acceleration is the equal change of the velocity at equal times. To demonstrate different acceleration values, the angle of the ground is increased or decreased. Each experiment is repeated 5 times to ensure its precision.

(a): Acceleration

(\Delta v): Change in velocity (final velocity - initial velocity)

(Delta t): Time interval (final time - initial time)

$$a = \frac{\Delta v}{\Delta t}$$



2. Figure 2: Setup for experiment of constant acceleration

3. Calculations

Part 1

Figure 1 : Setup for experiment of constant velocity

Table 1: Average velocities of each run

Run	Average Velocity
1	0.242
2	0.221
3	0.242
4	0.417
5	0.754

Related Equations:

$v = \Delta x / \Delta t$

theoretical result – experimental result = margin of error

error percentage = | margin of error – experimental result | * 100

Calculations:

$\Delta x = 0.559 – 0.192 = 0.367$
$\Delta t = 2.50 – 1.05 = 1.45$
$0.367 / 1.45 = 0.253$
$0.253 – 0.242 = 0.011$ margin of error
Error percentage = $|0.011 / 0.242| * 100 = 4.5\%$

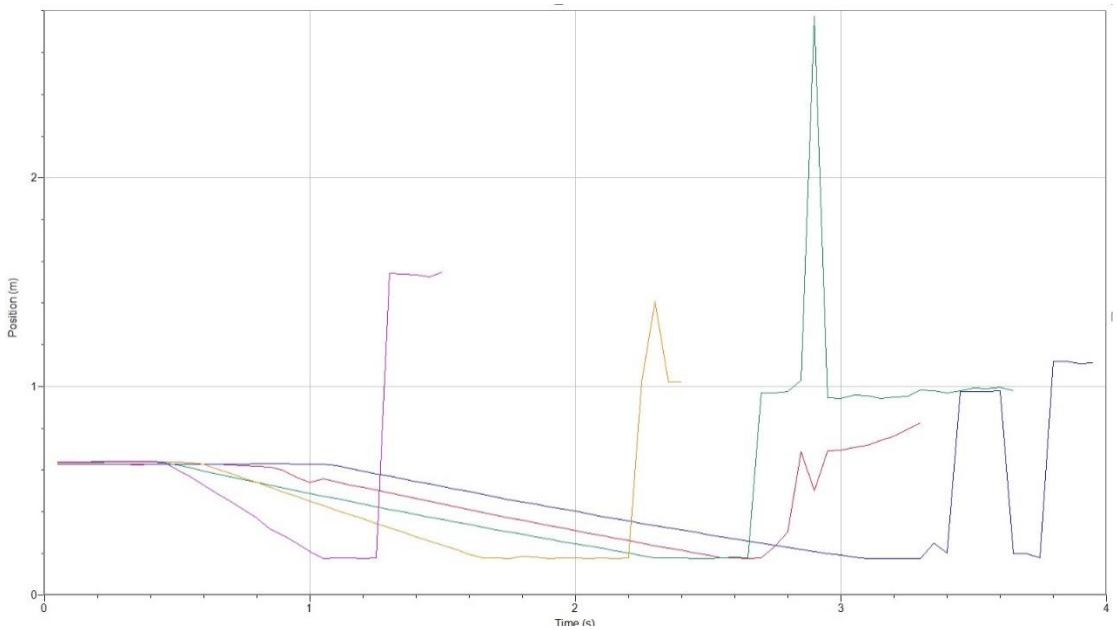
$\Delta x = 0.619 – 0.191 = 0.428$
$\Delta t = 3 – 1.10 = 1.9$
$0.428 / 1.9 = 0.197$
$0.197 – 0.221 = -0.024$ margin of error
Error percentage = $|0.024 / 0.221| * 100 = 10.8\%$

$\Delta x = 0.623 – 0.189 = 0.434$
$\Delta t = 2.25 – 0.50 = 1.75$
$0.434 / 1.75 = 0.248$
$0.248 – 0.242 = 0.006$ margin of error
Error percentage = $|0.006 / 0.242| * 100 = 2.4\%$

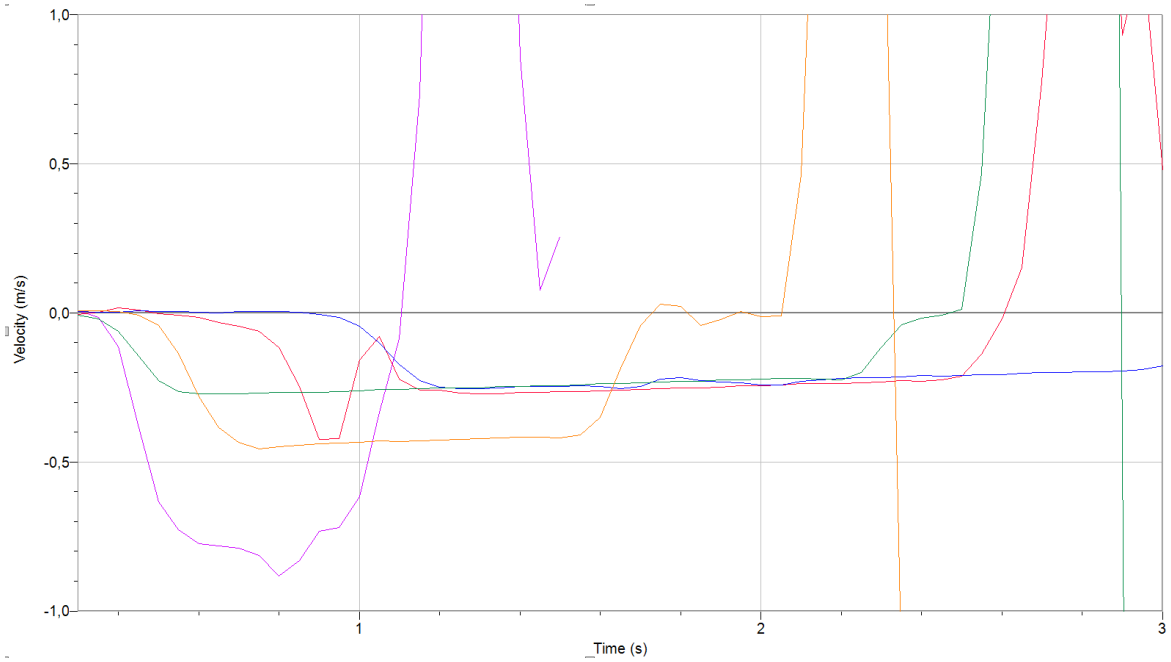
$\Delta x = 0.626 – 196 = 0.430$
$\Delta t = 1.60 – 0.60 = 1$
$0.430 / 1 = 0.430$
$0.430 – 0.417 = 0.013$ margin of error
Error percentage = $|0.013 / 0.417| * 100 = 3.1\%$

$\Delta x = 0.639 – 0.209 = 0.430$
$\Delta t = 1 – 0.45 = 0.55$
$0.430 / 0.55 = 0.781$
$0.781 – 0.754 = 0.027$ margin of error
Error percentage = $|0.027 / 0.754 | * 100 = 3.5\%$

Graphs :



Graph 1 : Position – Time graph of each run



Graph 2: Velocities of each run

Part 2

Table 2: Acceleration values of different inclination angles

Run	**$\Theta_{1=10} \pm 0.1$**	**$\Theta_{2=6} \pm 0.1$**	**$\Theta_{3=4} \pm 0.1$**
1	0.804	0.650	0.578
2	0.835	0.586	0.503
3	1.057	0.649	0.510
4	1.196	0.766	0.467
5	0.979	0.619	0.532

4. Conclusion

In this experiment, we throw objects to motion detector and measured constant velocity and constant acceleration. Then, we did theoretical calculations and we compared theoretical results and experimental results. According to the comparison, we calculate margin of error.

5. References

1. Physics 1103.4 2nd Lab Manual
- https://canvas.instructure.com/files/275439797/download?download_frd=1

2. University Physics with Modern Physics, Hugh D. Young, Roger A. Freedman, 15th Edition, Pearson, 2019.

6.Appendix

Experiment 2 Constant Velocity and Acceleration

Section Number: PHYS1103. ___ Date: __/__/____

Group Number: ___

	Dept	Student Number	Name	Surname	Task Distribution
1	CSE	150123046	Cihanem ~~Yamaç~~	YAMAÇ	
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3	CSE	150124060	Yusuf Gelil	Gelil	

1. Objectives

1) To examine constant velocity and acceleration motion.
2) To learn how to obtain equations from graphs using curve fitting.
3) To determine percentage errors between experimental and theoretical studies.
4) Be able to comment on errors

2. Equipment List

1) Vernier Labquest Mini
2) Motion Detector
3) Moving Object
4) Ruler

3. Set-up and Procedure:

This experiment has two parts. The constant speed experiment will be performed in the first part, and the constant acceleration experiment will be performed in the second part. The steps of these experiments are written in the procedures of each part. Before coming to the laboratory, you should write the equations related to the experiments in the "Related Equations" section of the experiment sheet.

PART 1. A. Constant Velocity in Straight Line Motion

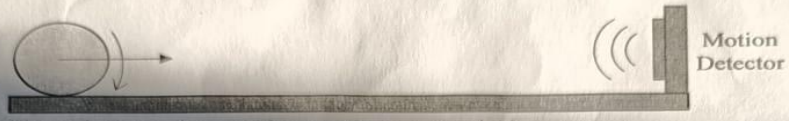


Figure 1. Setup for experiment of constant velocity

Procedure

1) Place the object in front of the motion detector
2) Start the Logger Lite then start the record.
3) Push and release the object to the motion detector.
4) Write the value of the velocity in Table 1 from the graph of the Logger Lite
5) Repeat the experiment 5 times.
6) Show plots in one graph

Table 1. Average velocities of each run

Run	Average Velocity
1	-0,242 m/s
2	-0,221 m/s
3	-0,242 m/s
4	-0,417 m/s
5	-0,754 m/s

Related Equations:

$x = v.t$

PART 2. Constant Acceleration in Straight Line Motion

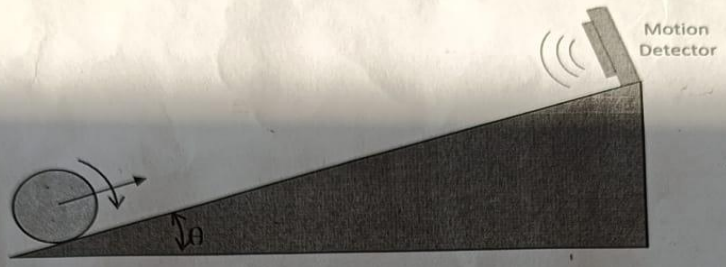


Figure 2. Setup for constant acceleration

Procedure

1) Set the angle of the inclination of the plate and record it in Table 2
2) Place the motion detector
3) Place the object near the bottom of the plate
4) Start the Logger Lite then start the record.
5) Push up the object and wait until returns to its initial position.
6) Write the value of the acceleration in Table 2 from graph of the Logger Lite
7) Repeat the experiment 3 times for 3 angles of the plate.
8) Plot the position-time and velocity-time graphs

Table 2. Acceleration values of different inclination angles

Run	$\theta_1 = 10 \pm 0,1$	$\theta_2 = 6 \pm 0,1$	$\theta_3 = 4 \pm 0,1$
1	0,804	0,650	0,578
2	0,835	0,586	0,503
3	1,057	0,649	0,510
4	1,196	0,766	0,467
5	0,979	0,619	0,532

Related Equations:

$\frac{\Delta v}{\Delta t} = a$ $\quad F_s = mg \cdot \cos\theta \cdot \mu$